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TSYP14

IEEE TUNISIAN STUDENT AND YOUNG PROFESSIONAL CONGRESS

IEEE CN CONSULTANTS NETWORK TUNISIA SECTION | IEEE COMPUTER SOCIETY Tunisia Chapter

City Brain Convergence

TSYP14 TECHNICAL CHALLENGE

SCOPE & TOPIC

Build a connected three-layer smart city system:

1. **TWIN LAYER**

Simulates the city's sensors and actuators.

2. **CITY BRAIN LAYER**

Consumes real-time data and makes operational decisions.

3. **GUARDIAN LAYER**

Monitors the same data to detect anomalies, malfunctions and cyber-attacks, and acts to contain them.

Teams are grouped into triplets — one team per layer — and must make the three systems interoperate as one functioning smart city.
Security and trust are core design requirements.

Challenge problem

Smart cities depend on sensing, deciding and trusting all working together, yet most systems are built to do only one well.
The challenge is to build all three as separate, independently developed systems and make them work as one, including when the data itself is being falsified.

Challenge Goal

Each team's goal is to:

- Build a working and correct system for its assigned layer:
  Twin, City Brain or Guardian
- Connect it with the two other layers built by its matched partner teams.
- Create a single functioning, smart and safe digital simulation of a city.
- Ensure the team's own layer works correctly in isolation.
- Ensure it interoperates with systems the team did not build.
- Operate under a shared technical contract.
- Test the integrated system under realistic and adversarial conditions.

The challenge emphasizes engineering to an interface you do not control and proving that your system remains reliable when connected to external systems.

The Three Layer

TWIN LAYER

Design and build the smart-city digital twin, the simulated environment where the other two layers connect.

Simulate city sensors:

- Traffic cameras
- Water-level sensors
- Power-load meters
- Air-quality monitors
- Emergency-call feeds

Simulate actuators:

- Police / ambulance dispatch
- Traffic-signal control
- Water-valve control
- Power-grid switching
- Device isolation / quarantine

Expose every sensor reading and actuator action as a callable MCP-style tool so City Brain and Guardian can read and act on them.

CITY BRAIN LAYER

Monitor the Twin's live sensor stream for real-world incidents such as:

- Accidents
- Fires
- Traffic jams
- Flooding

The City Brain must:

- Decide the correct operational response.
- Dispatch the matching actuator through the Twin's tools.

GUARDIAN LAYER

Monitor the same sensor stream as City Brain, but ask a different question:
"Is this sensor telling the truth?"
Detect:

- Sensor malfunction
- Actuator failure
- Spoofed data
- Cyber-attacks

Contain confirmed problems by:

- Isolating a compromised sensor
- Quarantining a device
- Rolling back a falsified reading

Provide City Brain with a trust verdict on a reading before City Brain is allowed to act on it.

Challenge instructions

Teams are organized into triplets, with one team assigned to each layer.
The three systems must ultimately operate together as one functioning smart-city system.

Phase 0: Ideation

01 September 2026 → 17 September 2026

Deliverable:

10-page PDF Report that must include:

- Proposal for the 3 layers
- Technical roadmap
- Technical choices
- Security measures

Team - Layer Assignment

Sep 22, 2026

- Assign Team A to Layer X
- Assign Team B to Layer Y
- Assign Team C to Layer Z

Detailed Specifications will be shared

Phase 1: Development

22 September 2026 → 14 November 2026

Teams Matching (Round 1)

31 October 2026

Match RANDOMLY Team A, Team B and Team C (These triplets will compete together until the CSTAM)

Triplets Demonstrations (Round 1)

14 November 2026 at CSTAM

Present and connect the matched teams' systems
Few Triplets will be selected for the next round

Working Prototype

Each team must have:

- A working prototype
- Individual evaluation

Established Connection

Each triplet must:

- Merge their systems
- Live-demonstrate the combined systems
- Undergo triplet systems merge evaluation
- 20-page PDF Report that must include:
  - Technical documentation of the 3-layer implementation
  - Security measures
  - Technical-choice argumentation

Phase 2: Improvement

14 November 2026 → 14 December 2026

Teams Matching (Round 2)

1st December 2026

Match RANDOMLY Team A, Team B and Team C (These triplets will compete together until the end of the challenge)

Working Prototype

Each team must have:

- A working final prototype
- Individual evaluation

Established Connection

Each triplet must:

- Merge their systems
- Live-demonstrate the combined systems
- Test the system response to simulated scenarios provided by the challenge organizers
- Undergo triplet systems merge evaluation

20-page PDF Report that must include:

- Technical documentation of the 3-layer implementation
- Security measures
- Technical-choice argumentation

GitHub Repository that must contain:

- A well-documented and clear repository of the 3-layer system
- Necessary .md files for documentation

Team requirements

Maximum 4 members per team

Teams are grouped into triplets

Each triplet contains:

- 1 Twin team
- 1 City Brain team
- 1 Guardian team

Each team develops and owns one layer.

Deliverables

PHASE 0

- 10-page PDF Report

PHASE 1

- Working Prototype
- Integrated Triplet Demonstration
- 20-page PDF Report

PHASE 2

- Working Final Prototype
- Integrated Triplet Demonstration
- 20-page PDF Report
- GitHub Repository

Scoring

TOTAL SCORE: 100 POINTS

Phase 1: 50 pts

- Individual Prototype Functionality: 10 pts
- Technical Implementation & Features: 5 pts
- Interoperability & Connectivity Between the 3 Teams: 15 pts
- Security & Trust Mechanisms: 10 pts
- Documentation & Code Quality: 5 pts
- Demonstration & Presence: 5 pts

Phase 2: 45 pts

- Final Prototype Functionality: 10 pts
- 3-Layer System Integration & Reliability: 10 pts
- Response to Adversarial/Simulated Scenarios: 15 pts
- Final Presentation & Demonstration: 5 pts
- Technical Documentation & Repository: 5 pts

BONUS: 5 pts

- Having at least 1 CS member: 1 pt
- Having at least 1 YP member: 1 pt
- Having at least 1 CN member: 1 pt
- Joint CyberSecurity LG Workspace on Collabrec: 0.25 pt per member (maximum points 1)
- Joint Artificial Intelligence LG Workspace on Collabrec: 0.25 pt per member (maximum points 1)

Pitching

English

Important dates

- Challenge Infosession: **09/09/2026**
- Phase 0 Deadline: **17/09/2026**
- Team-Layer Assignment: **22/09/2026**
- Teams Matching (Round 1): **31/10/2026**
- Phase 1 Deadline: **12/11/2026**
- Phase 1 Pitching: **14/11/2026 during CSTAM (details will be shared later)**
- Teams Matching (Round 1): **01/12/2026**
- Final Submission Deadline: **11/12/2026**

Winner

3 WINNING TEAMS

Prizes: To be determined

Submission

Submit your project and show us what you've built.

Contact



Please contact us through the following email address yp@ieee.tn or cs@ieee.tn or cn@ieee.tn for any inquires or clarifications.